

1 ShanghaiTech Crowd Counting Dataset

The ShanghaiTech Crowd Counting Dataset[1] is one of the most widely used benchmarks for crowd counting, density estimation, and crowd localization in computer vision. It was introduced in the influential paper Single Image Crowd Counting via Multi Column Convolutional Neural Network (MCNN)[2] and has become a standard dataset for evaluating crowd counting algorithms.

Each person in an image is labeled with a single dot near the center of the head. These point annotations are typically converted into density maps using Gaussian kernels during training.

1.1 Part A (ShanghaiTech-A)

Table 1: Statistics of the ShanghaiTech Part A Dataset

Property	Value
Images	482
Train	300
Test	182
Total People	~241,677
Average Count/Image	~501
Maximum Count	3,139
Minimum Count	33

Characteristics:

- Collected from the Internet.
- Extremely dense crowd scenes.
- Large scale variation.
- Severe occlusion.
- Strong perspective distortion.

1.2 Part B (ShanghaiTech-B)

Table 2: Statistics of the ShanghaiTech Part B Dataset

Property	Value
Images	716
Train	400
Test	316
Total People	~88,488
Average Count/Image	~124
Maximum Count	578
Minimum Count	9
Resolution	Mostly 768×1024

Characteristics:

- Collected from busy streets in Shanghai.
- Lower crowd density compared to Part A.
- More regular surveillance style viewpoints.
- Less severe occlusion.

1.3 Annotation Format

The original annotations are stored as MATLAB .mat files. For each image, there is a corresponding annotation file. It file contains a set of head-center coordinates:

$$[(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)]$$

where each coordinate represents the center of a person's head.

Example:

$$[(120, 45), (230, 89), (450, 150)]$$

This indicates that the image contains three annotated individuals. The crowd count is obtained by simply counting the number of annotation points.

2 Dataset References

Table 3: Representative Deep Learning-Based Crowd Counting Methods

Year	Paper	Model	Venue
2016	Single-Image Crowd Counting via Multi-Column CNN (MCNN)	MCNN	CVPR
2016	Fully Convolutional Crowd Counting on Highly Congested Scenes	FCN	arXiv
2017	Crowd Counting via Scale-Adaptive CNN (SaCNN)	SaCNN	CVPR
2017	Image Crowd Counting Using CNN and Markov Random Field	CNN-MRF	arXiv
2018	CSRNet: Dilated Convolutional Neural Networks for Understanding the Highly Congested Scenes	CSRNet	CVPR
2019	Crowd Counting with Deep Structured Scale Integration Network	DSSINet	CVPR
2019	CAN: Context-Aware Network for Crowd Counting	CAN	CVPR
2019	MRCNet: Crowd Counting and Density Map Estimation in Aerial and Ground Imagery	MRCNet	Remote Sensing
2020	BL: Bayesian Loss for Crowd Count Estimation	Bayesian Loss	ICCV
2020	DM-Count: Distribution Matching for Crowd Counting	DM-Count	CVPR
2021	SASNet: Scale-Aware Selection Network	SASNet	AAAI
2021	Deep Fusion Network for Crowd Counting in High-Density Scenes	D-Fusion	Journal
2022	CCTrans: Crowd Counting Transformer	Transformer	BMVC
2023	PET: Point Query Transformer	Transformer	ICCV
2024	Label Noise Robust Crowd Counting with Loss Filtering	Loss Filtering	Journal

References

- [1] L. Zhu, C. Li, Y. Zhongguo, K. Yuan, and S. Wang, “Crowd density estimation based on classification activation map and patch density level,” *Neural Computing and Applications*, vol. 32, 05 2020.
- [2] Y. Zhang, D. Zhou, S. Chen, S. Gao, and Y. Ma, “Single-image crowd counting via multi-column convolutional neural network,” in *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 2016, p. 589–597. [Online]. Available: <http://dx.doi.org/10.1109/CVPR.2016.70>